

Cell Injury and adaptation (I)

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PA0097, PA0313

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Case List

Cell Injury and adaptation (I)

No.			
PA0097	(77)	Kidney	Acute tubular necrosis
PA0313	(196)	Colon	Necrotizing enterocolitis
PA0096	(76)	Kidney	(1) Infarction (2) Acute tubular necrosis

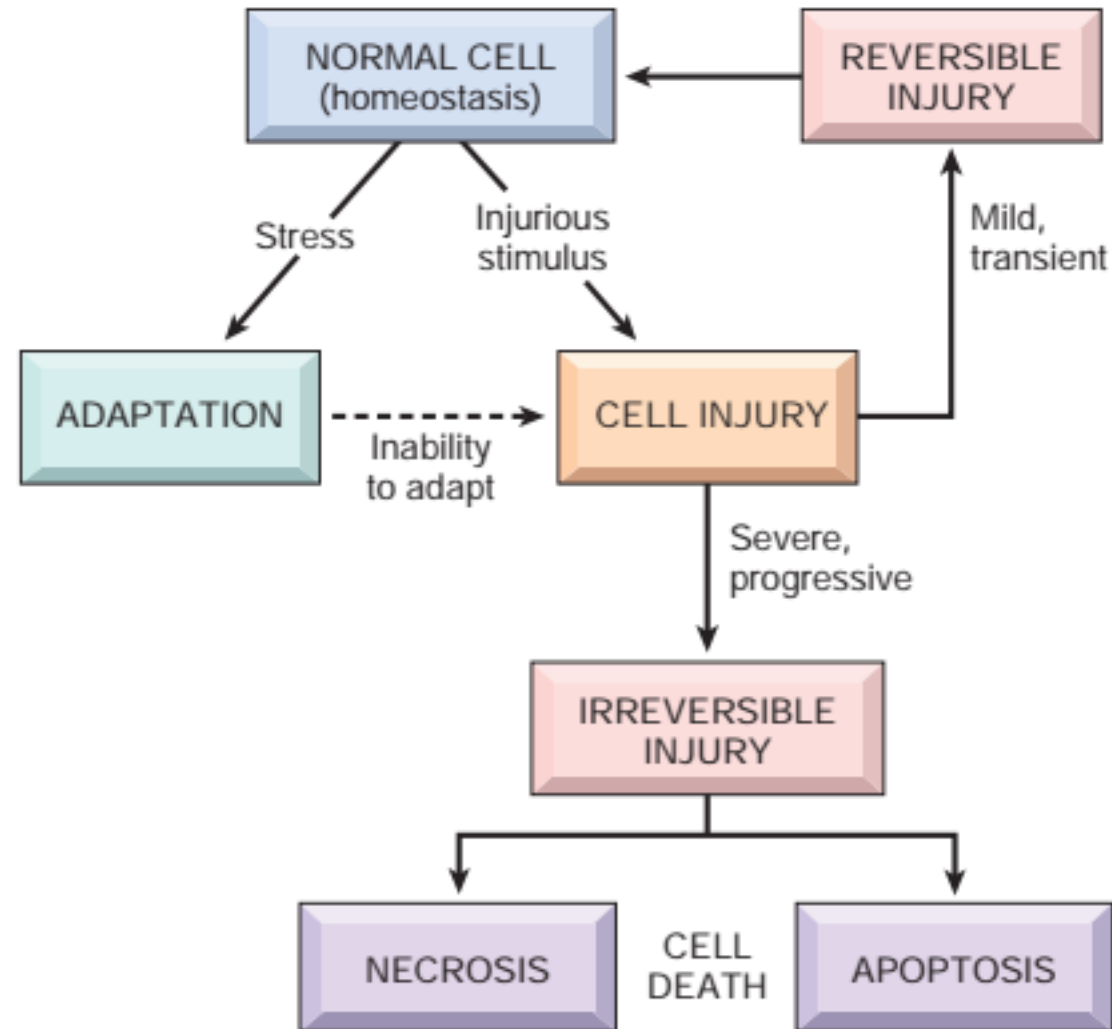
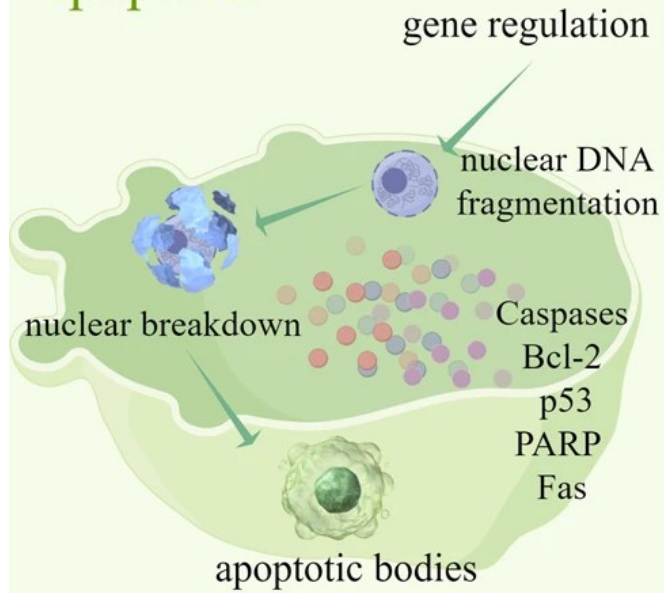
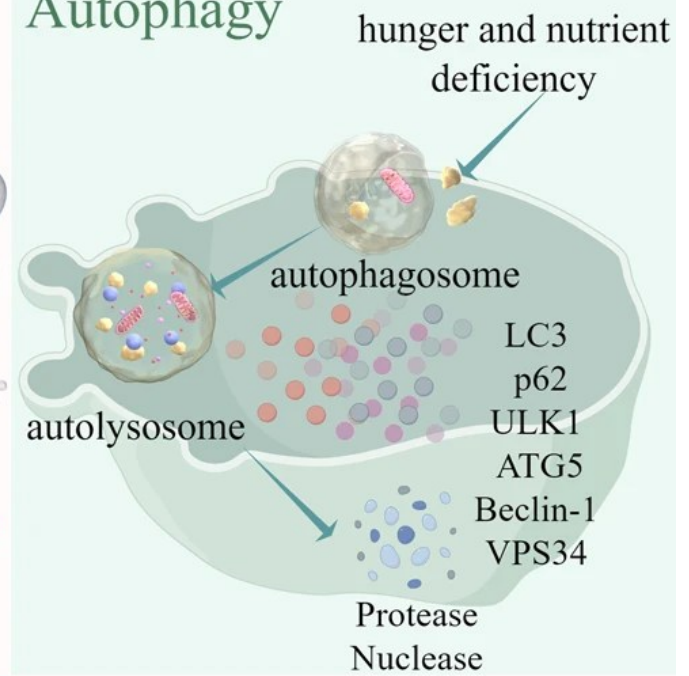


Figure 2.1 Stages of the cellular response to stress and injurious stimuli.

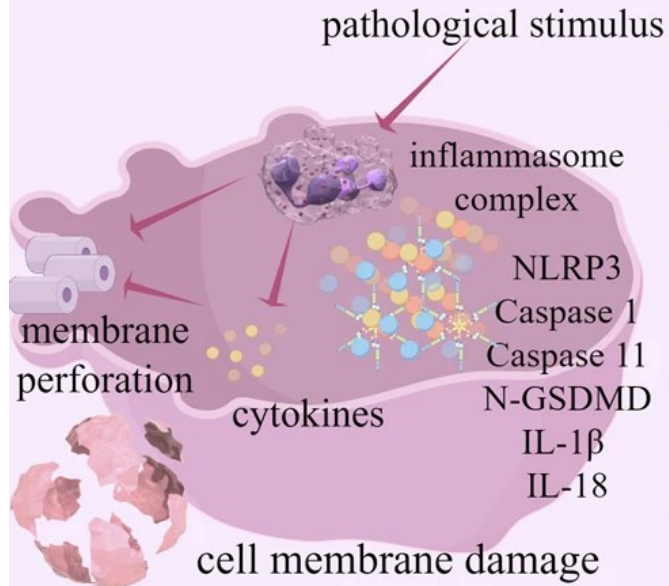
Apoptosis



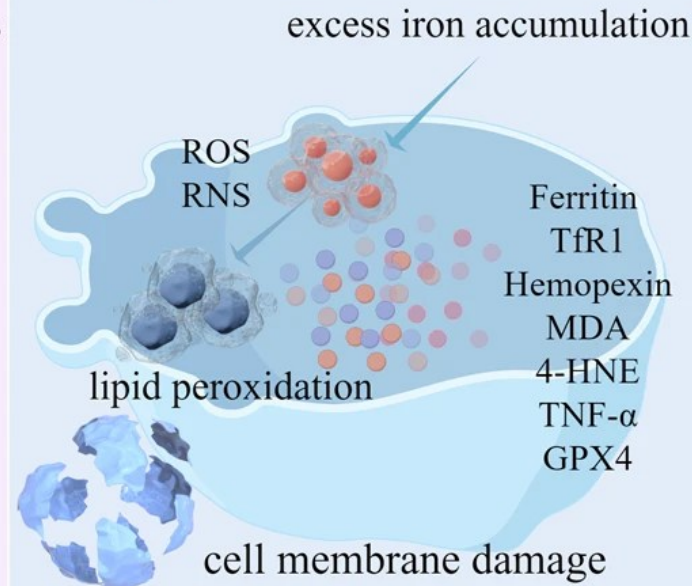
Autophagy



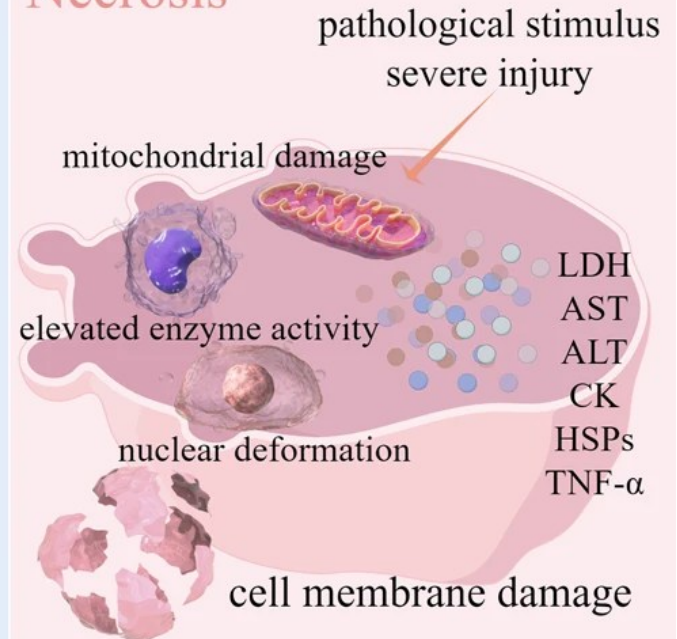
Pyroptosis



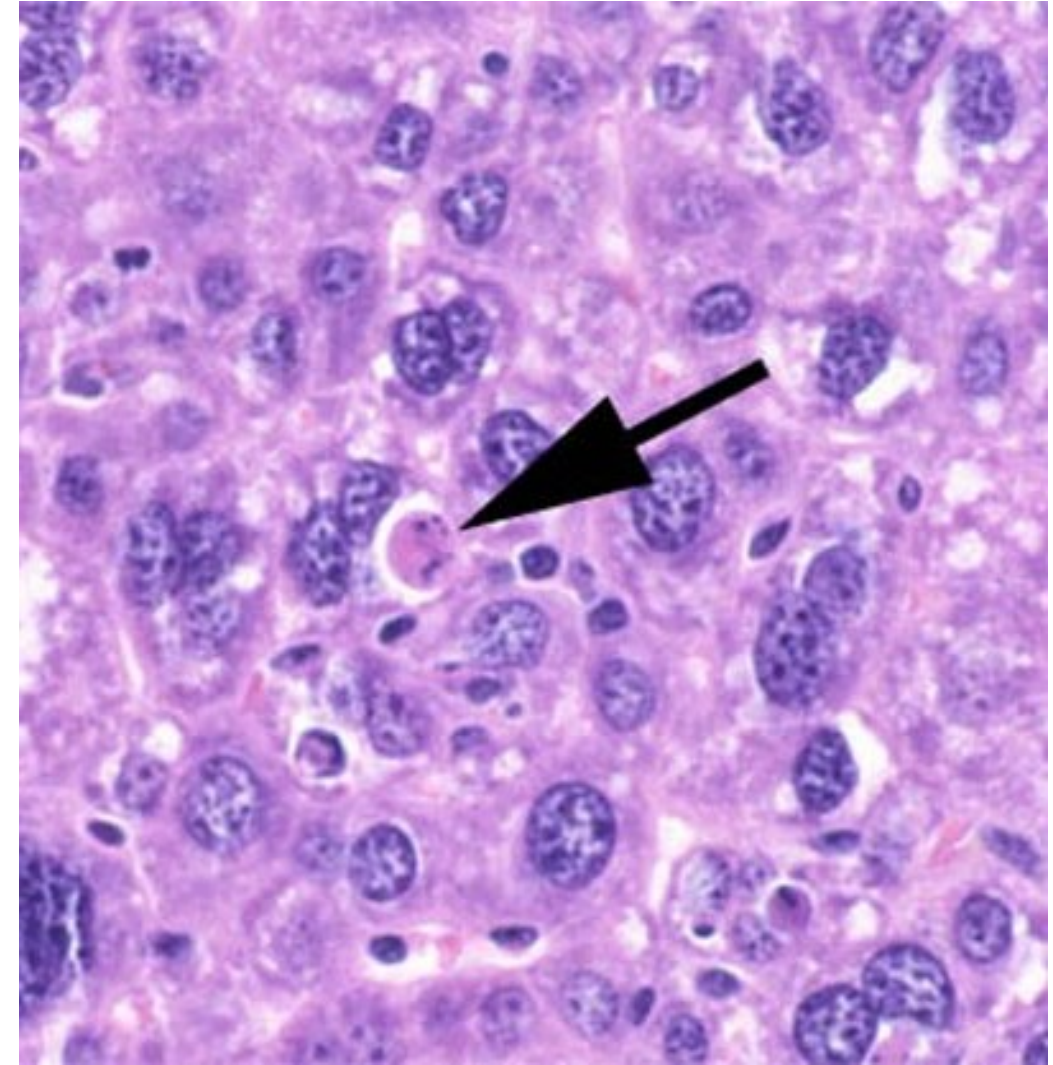
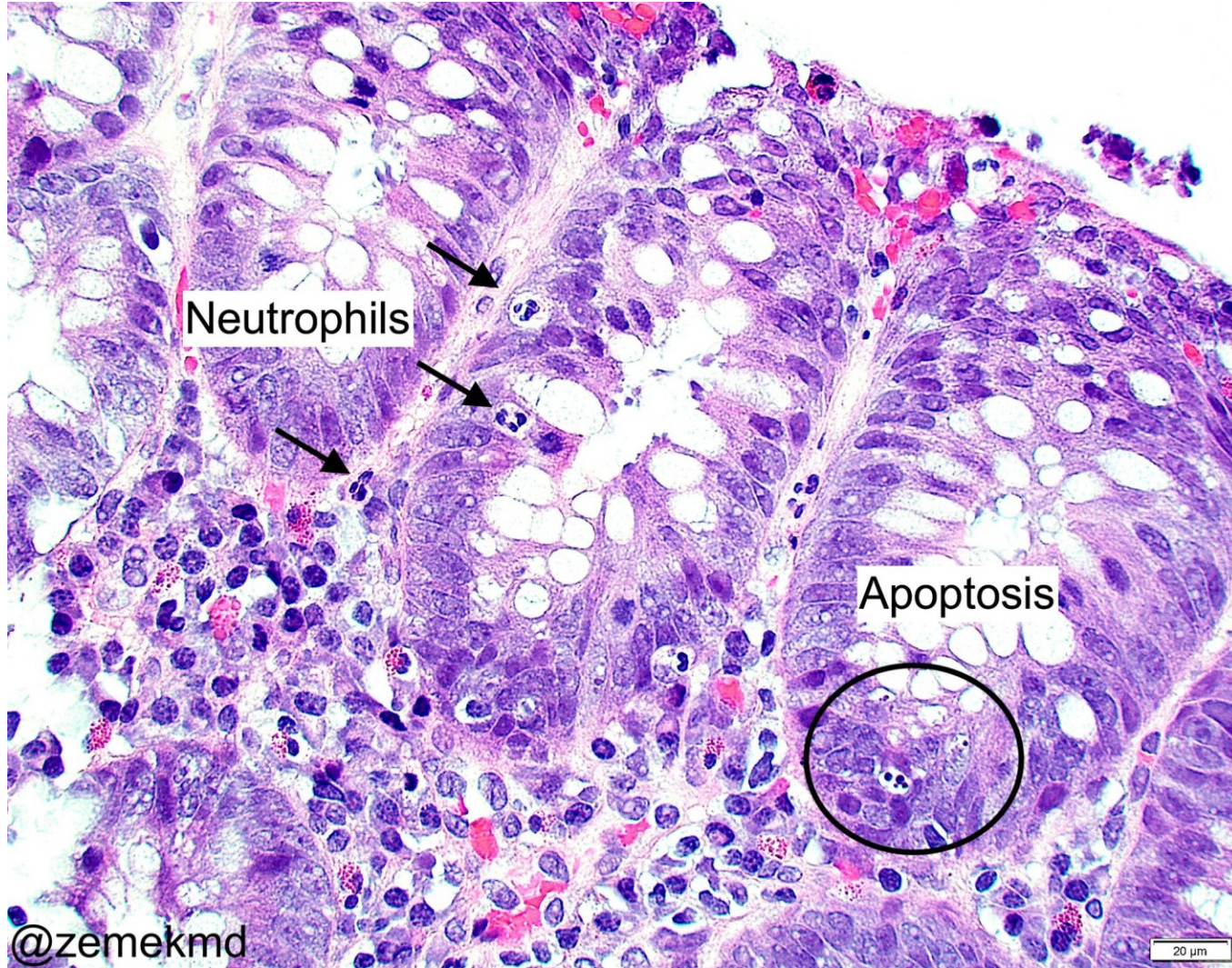
Ferroptosis



Necrosis



Apoptosis (非本堂課重點)

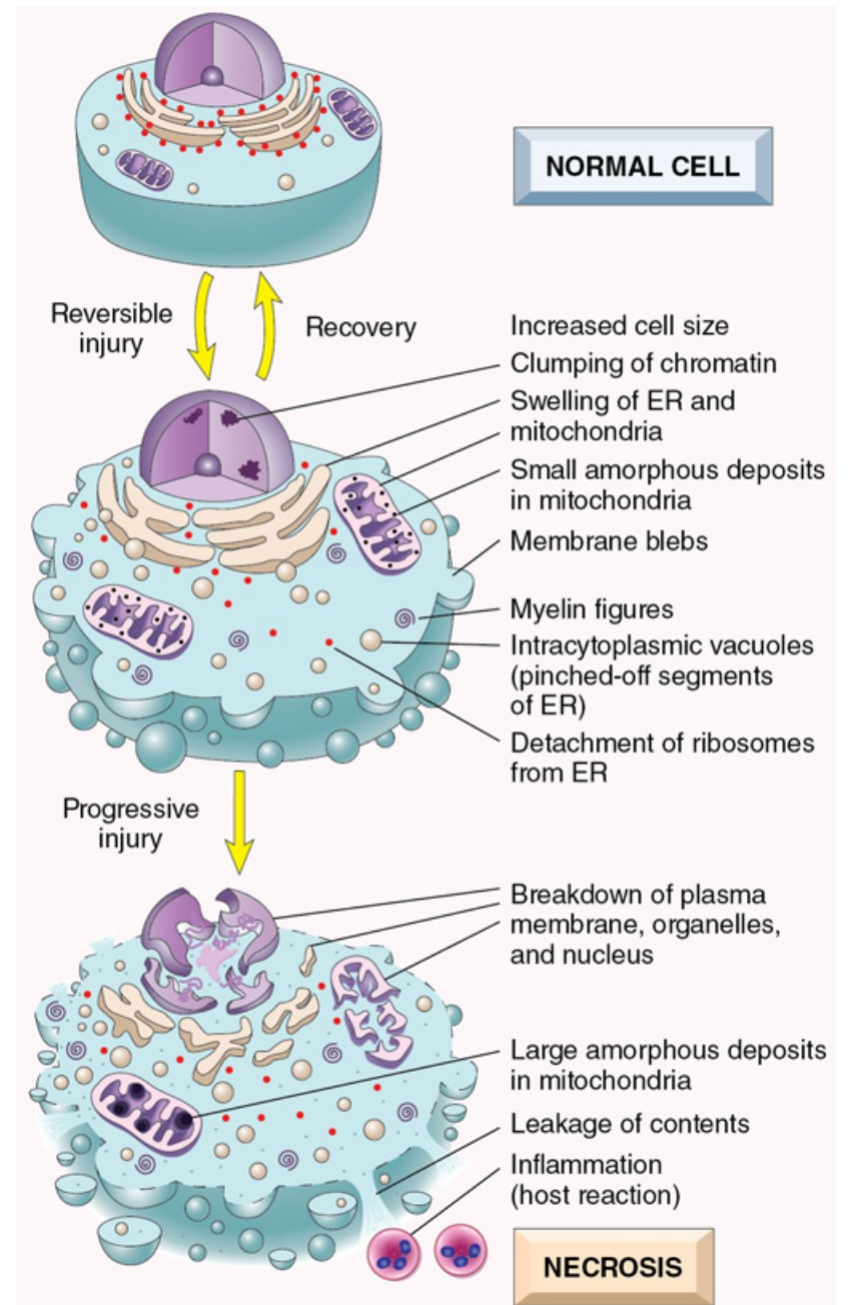


Necrosis

- Definition
 - A pathologic process that is the consequence of severe injury
- Mechanism
 - Denaturation of cellular proteins
 - Leakage of cellular contents through damaged membranes
 - Local inflammation
 - Enzymatic digestion of the lethally injured cell

Morphology

- Cell injury
 - Cellular swelling
 - Hydropic change/vacuolar degeneration
- Necrosis
 - Cytoplasmic eosinophilia
 - Attributable to loss of RNA & ↑ denatured cytoplasmic protein
 - Nuclear changes: karyolysis (loss of nuclei), pyknosis (shrinkage), karyorrhexis (fragmentation)



Necrosis

- Patterns
 - **Coagulative necrosis (PA0097, PA0313)**
 - Liquefactive necrosis (PA0162)
 - Caseous necrosis (PA0198)
 - Fat necrosis (PA0081)
 - Gangrenous necrosis
 - Fibrinoid necrosis

Coagulative Necrosis 凝固性壞死

- Definition
 - A form of necrosis in which the architecture of dead tissue is preserved for a span of at least some days
- Cause
 - Ischemia → Coagulative necrosis of the supplied tissue in all organs except brain
- Mechanism
 - The injury denatures not only structural proteins but also enzymes and so blocks the proteolysis of the dead cells
 - Ultimately, the necrotic cells are broken down by lysosomal enzymes from infiltrating leukocytes

Coagulative Necrosis - Morphology

Gross

- Firm texture → “凝固性”

“Ghost cells”

Microscopic

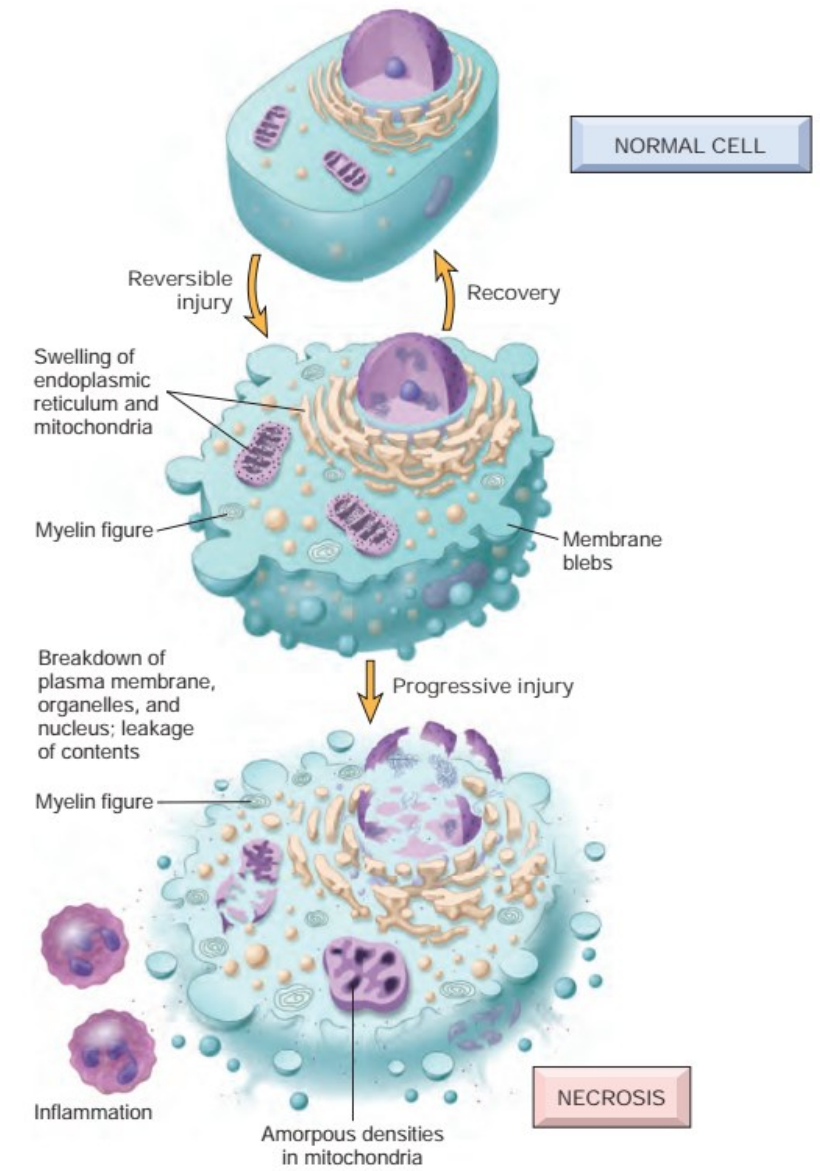
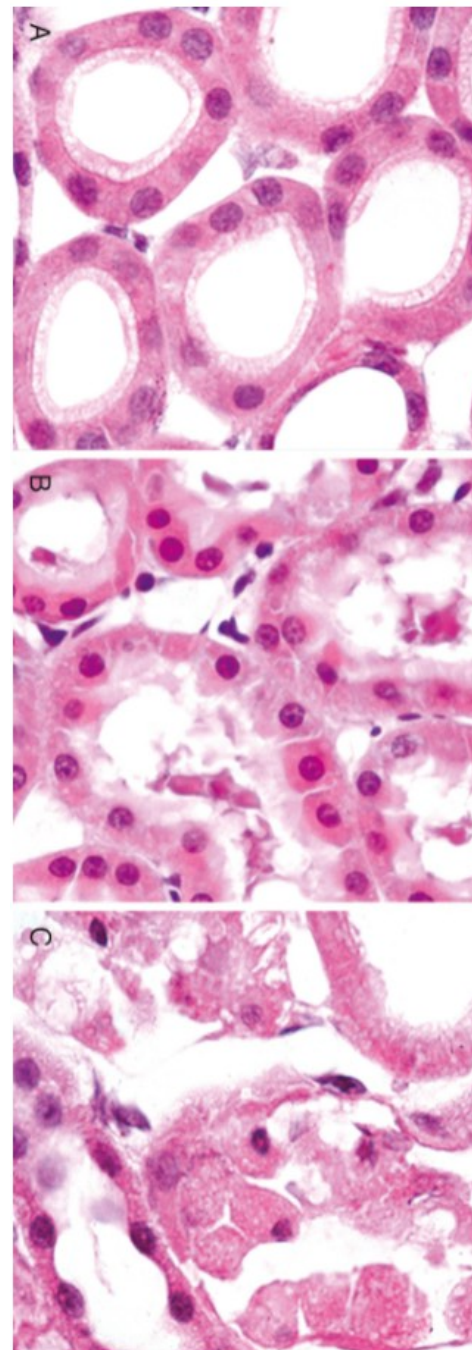
- **Preserved cellular outlines**
- Cytoplasmic eosinophilia
- Nuclear changes: karyolysis, pyknosis, karyorrhexis
- An Inflammatory infiltrate

PA0097 (77)

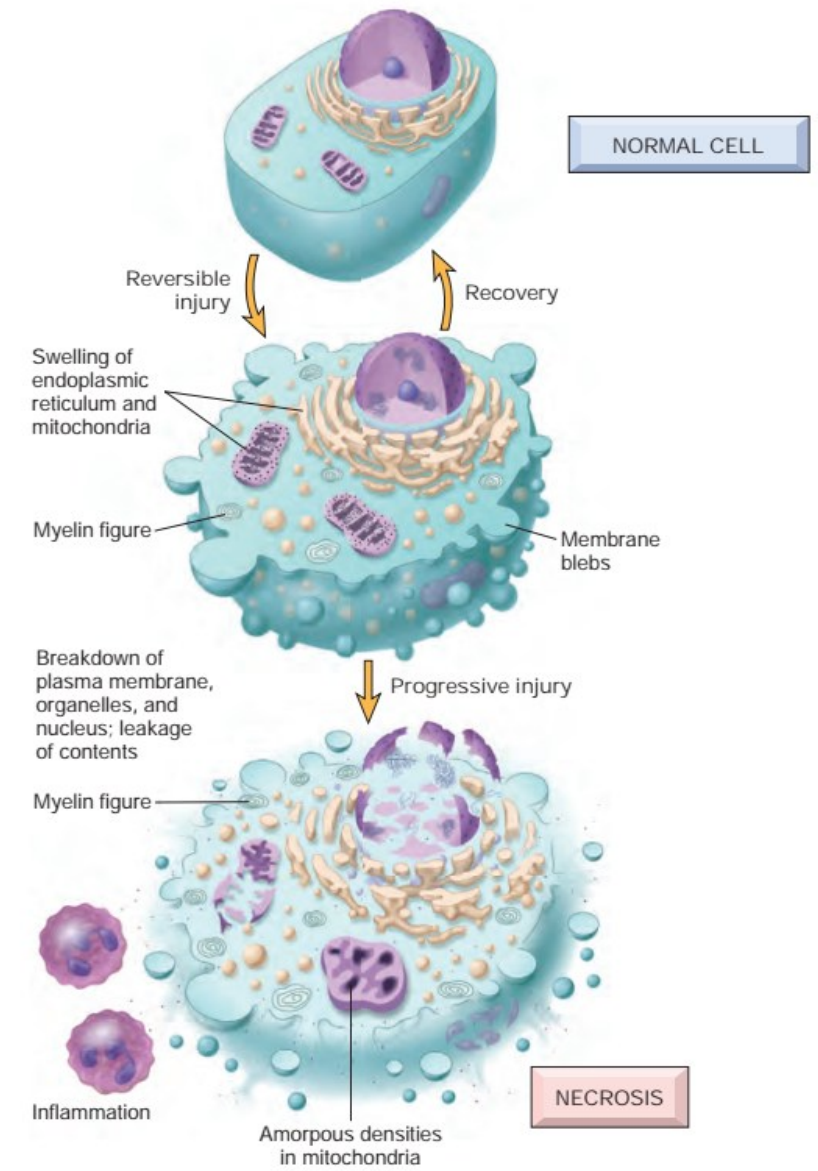
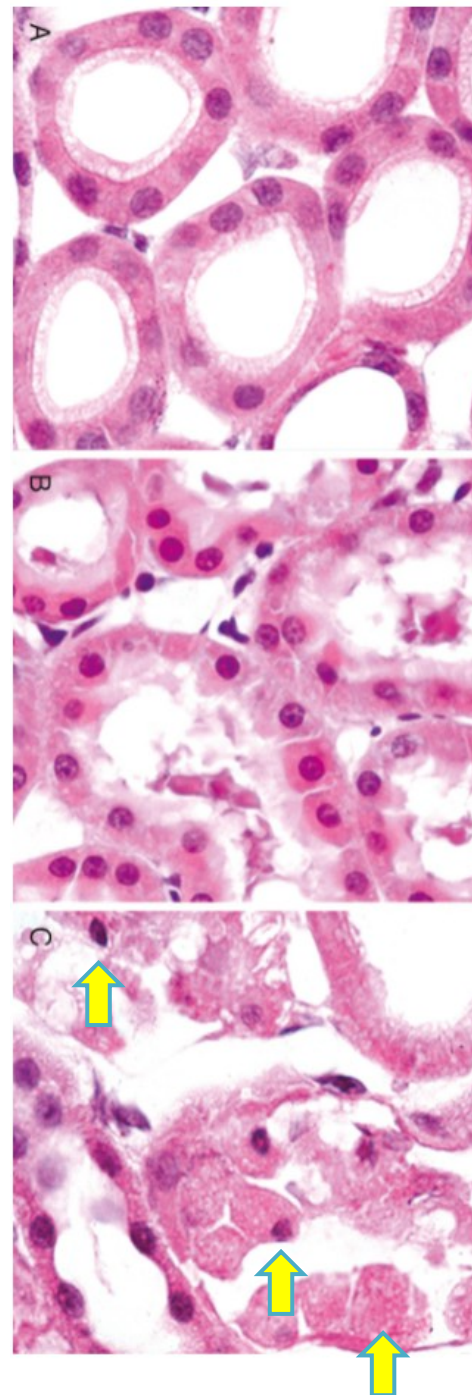
Acute tubular necrosis, kidney

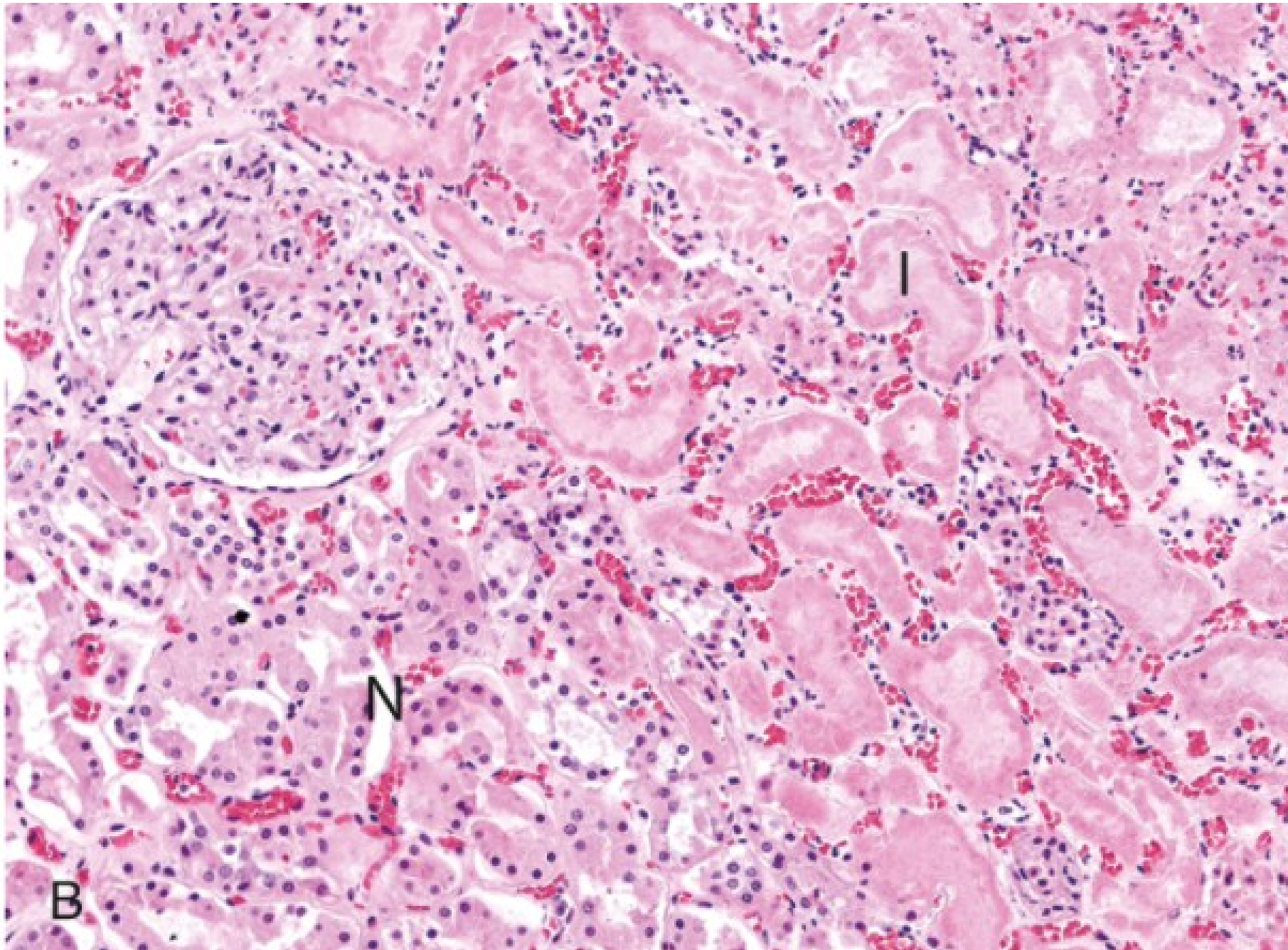
Robbins 11e, p36-40, p847-849

- Cell injury
 - Cellular swelling
 - Hydropic change/vacuolar degeneration



- Necrosis
 - Cytoplasmic eosinophilia
 - Nuclear changes:
 - karyolysis (loss of nuclei),
 - pyknosis (shrinkage),
 - karyorrhexis (fragmentation)





Infarct = A localized area of coagulative necrosis

Low power:

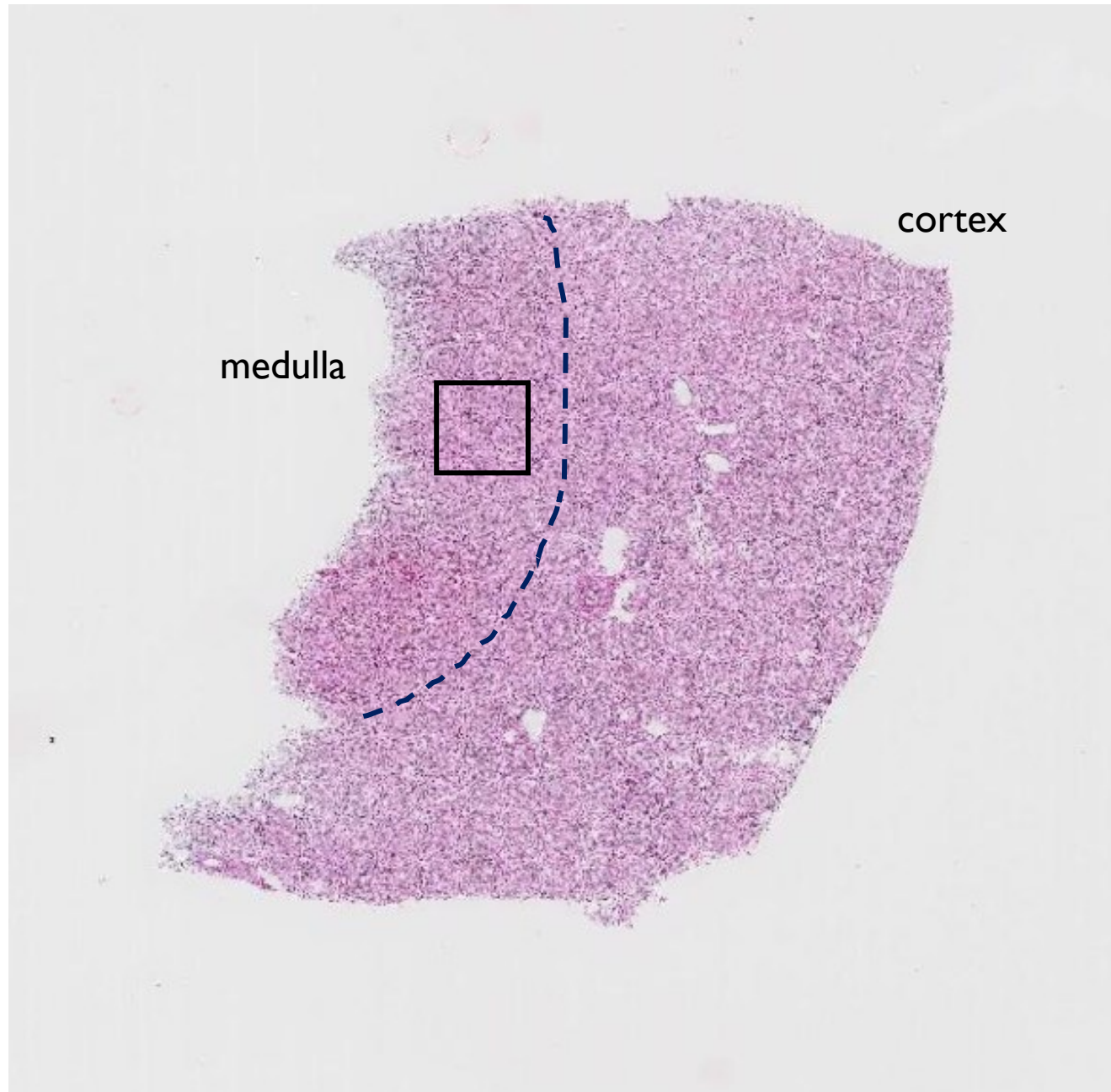
Clear border between normal & lesional parts

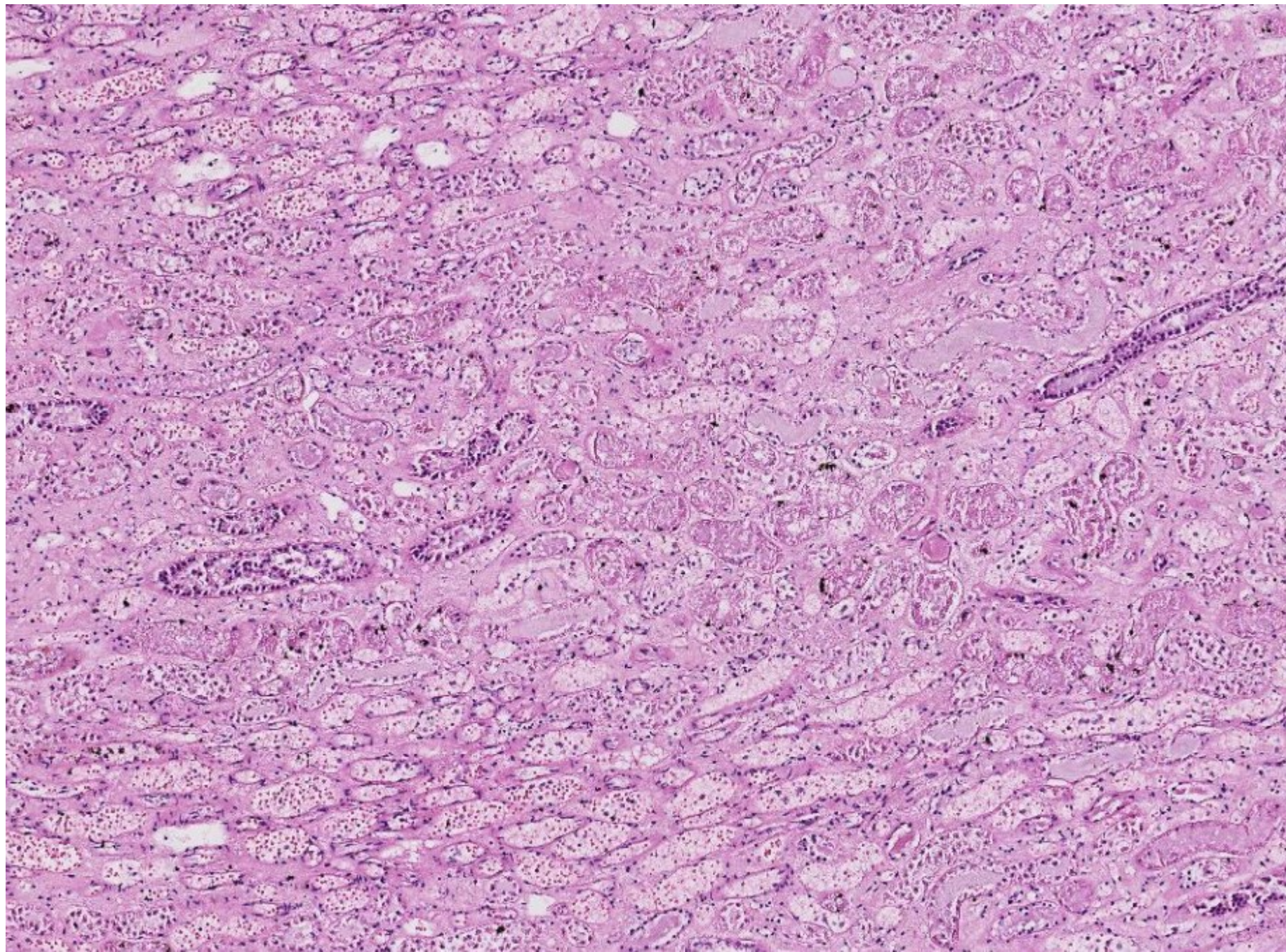
Preserved cellular outlines

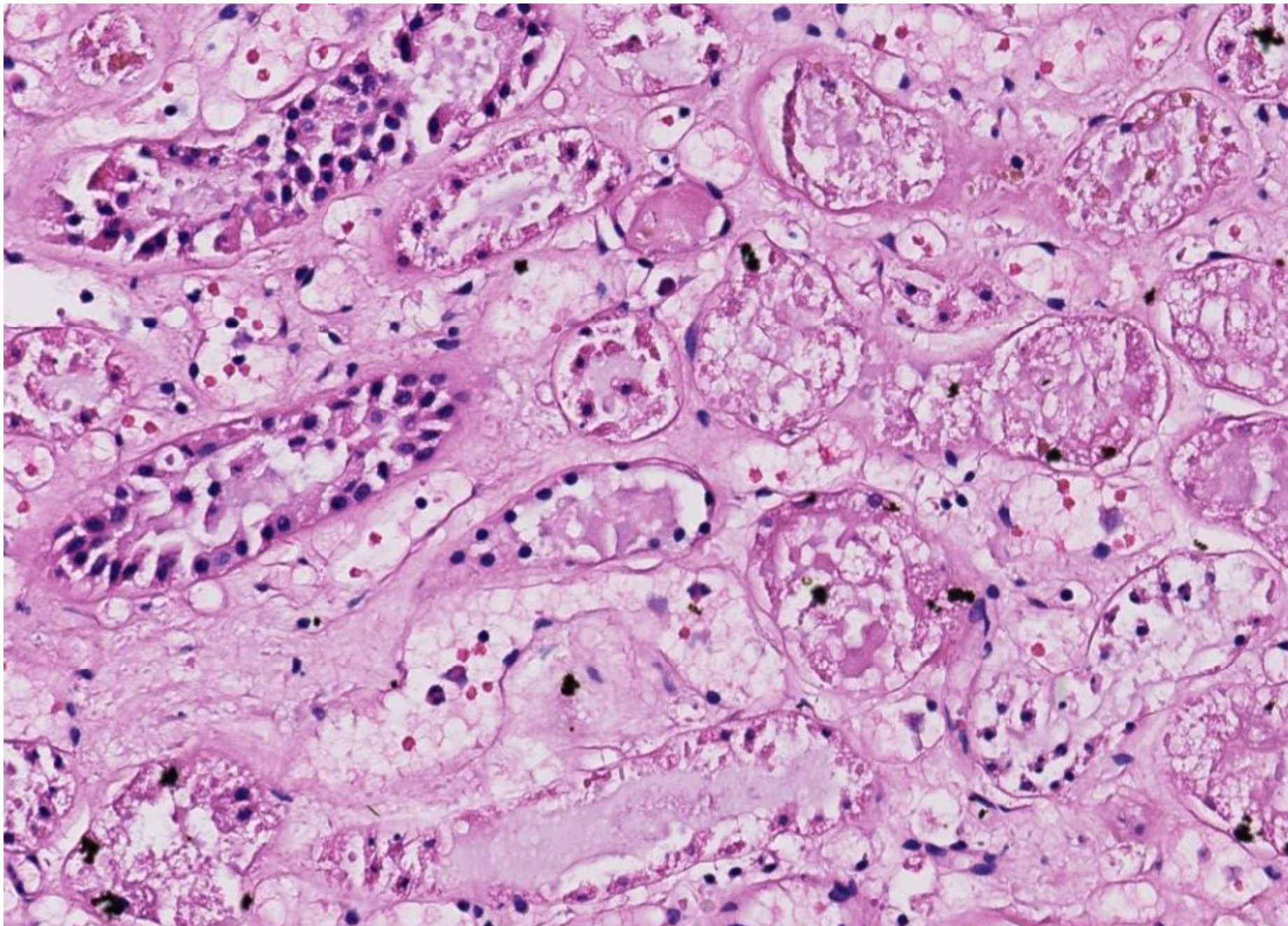
Cytoplasmic eosinophilia

Loss of nuclei

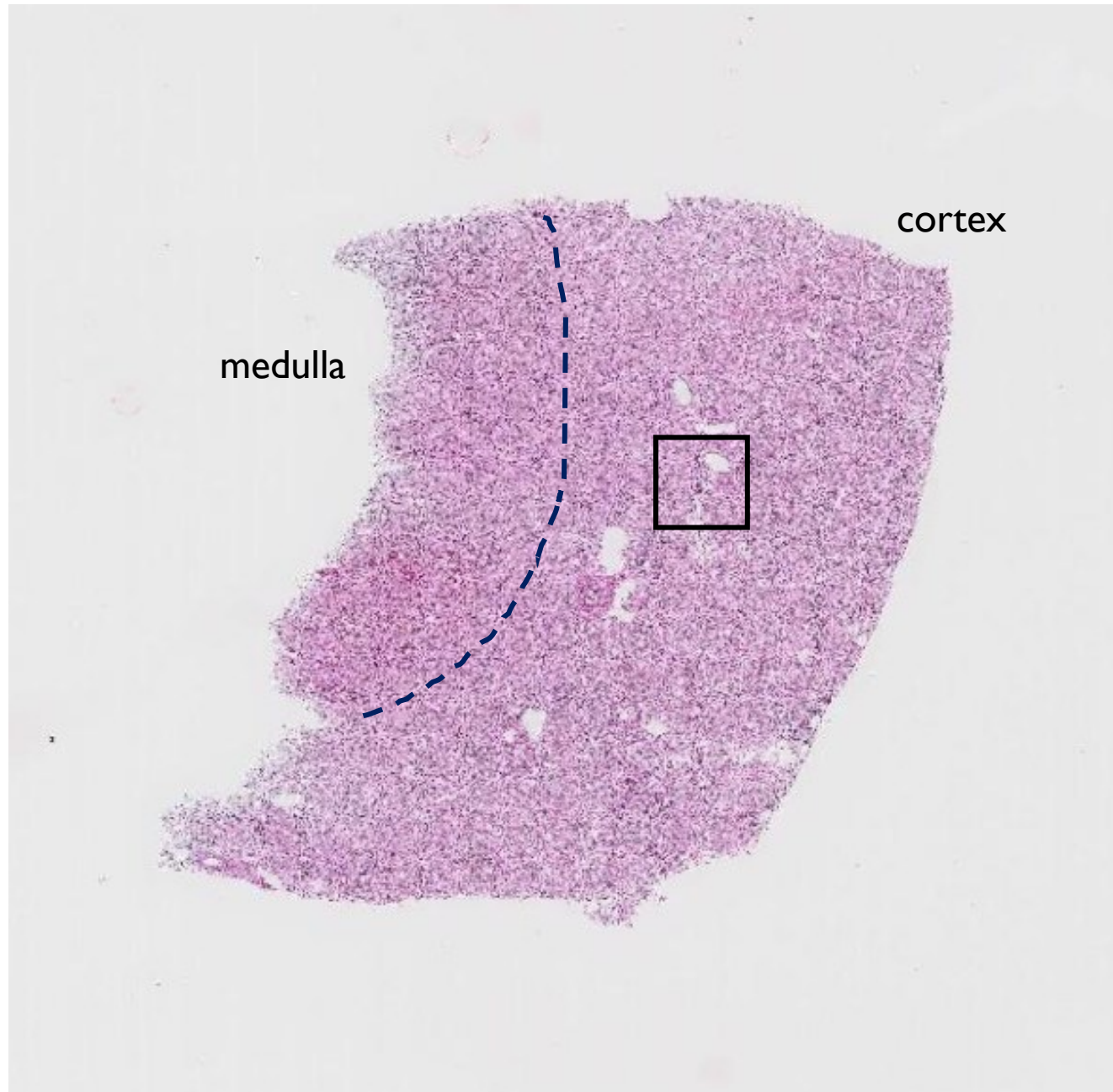
→ “Ghost cells”

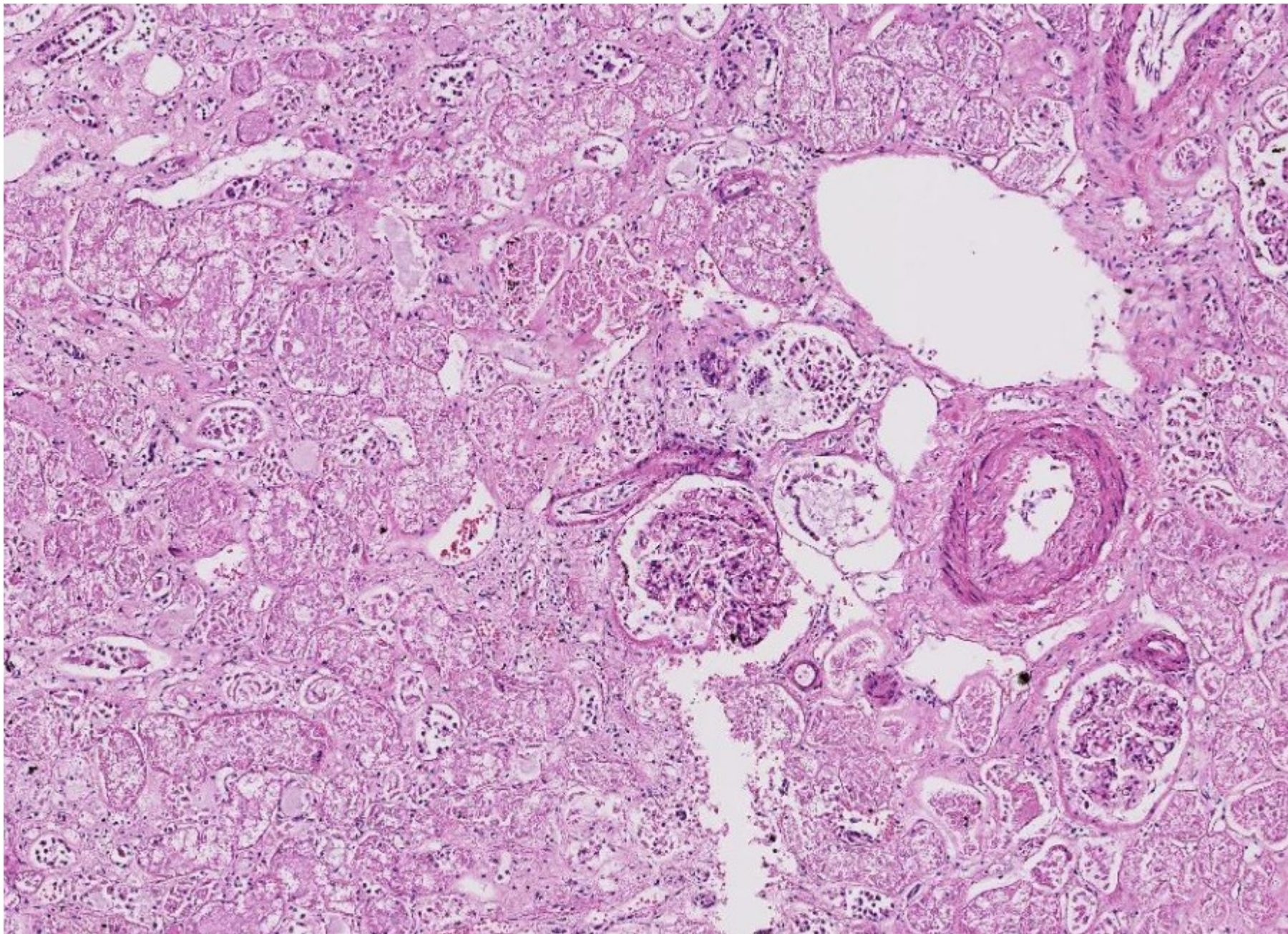


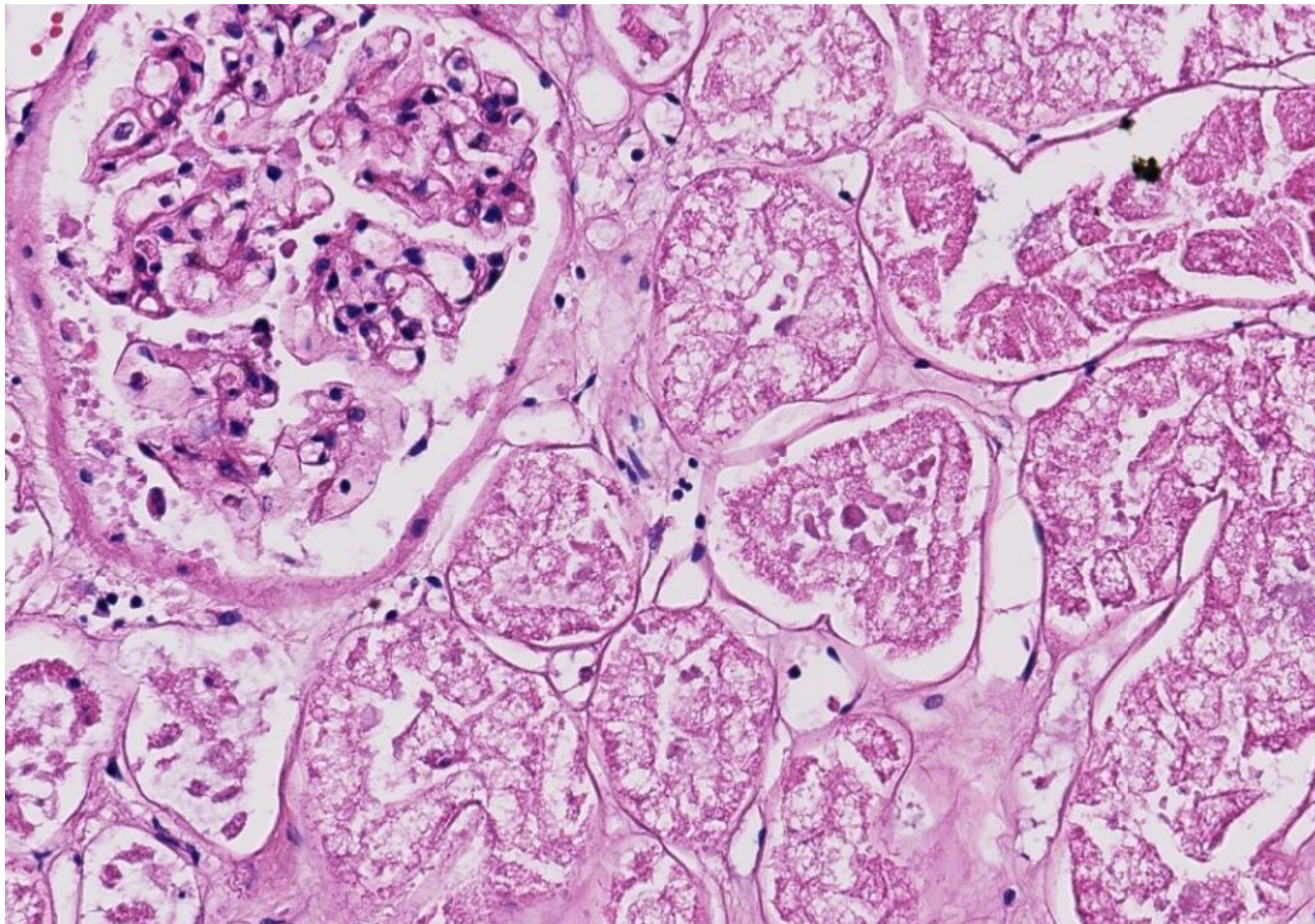




PA0097 (microteaching.ntu.edu.tw)







器官 (Organ)	Kidney
診斷 (Diagnosis)	Acute tubular necrosis
描述 (Description)	1. Tubular epithelial coagulative necrosis ① Dead cells with preserved cellular outlines ② Cytoplasmic eosinophilia, may glassy appearance, may vacuolated ③ Nuclear changes: karyolysis, pyknosis, karyorrhexis 2. An Inflammatory infiltrate

*Acute tubular injury/necrosis

mainly in proximal convoluted tubule(本片distal也很嚴重)

PA0313 (196)

Necrotizing enterocolitis

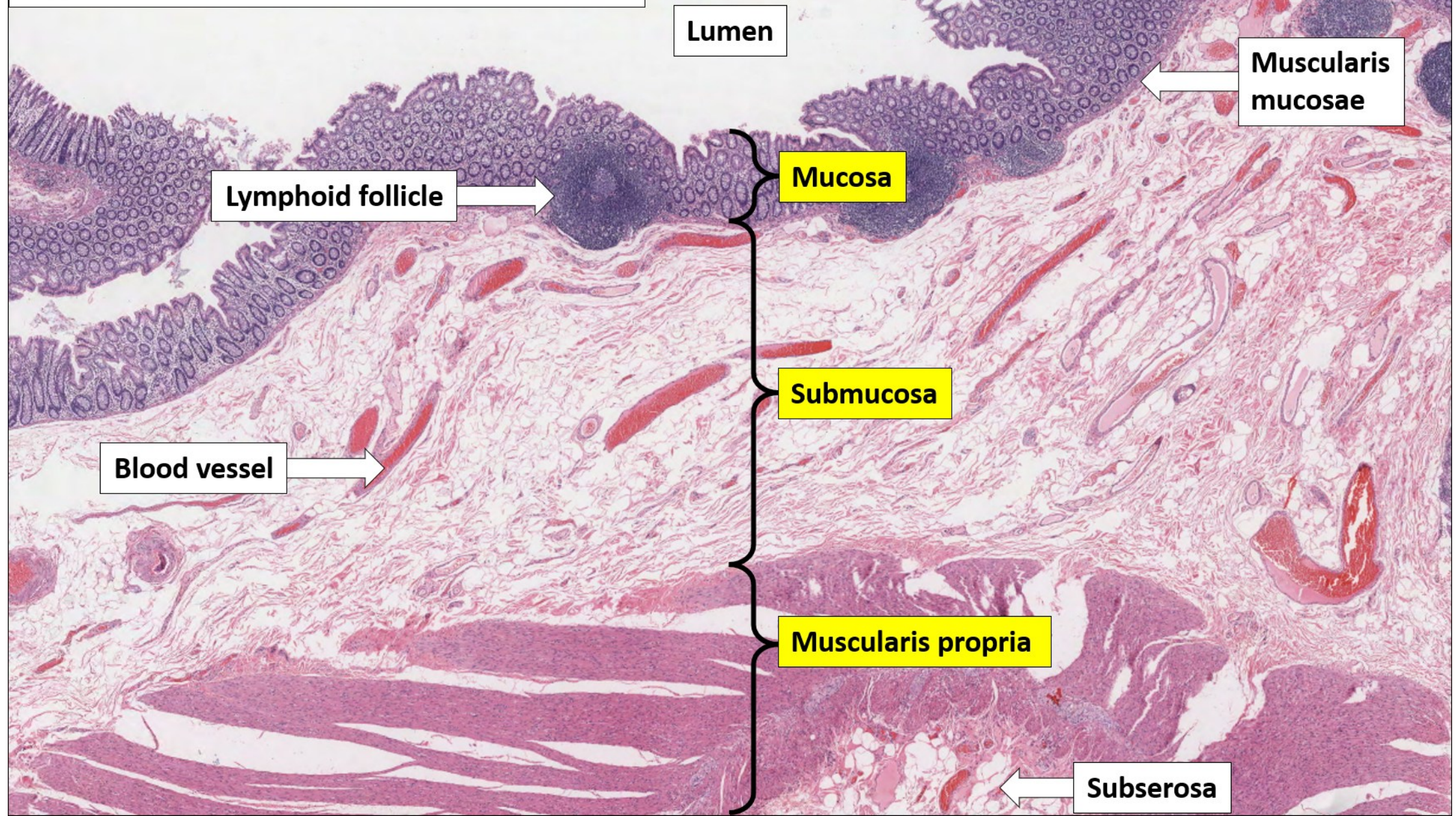
Robbins 11e, p418

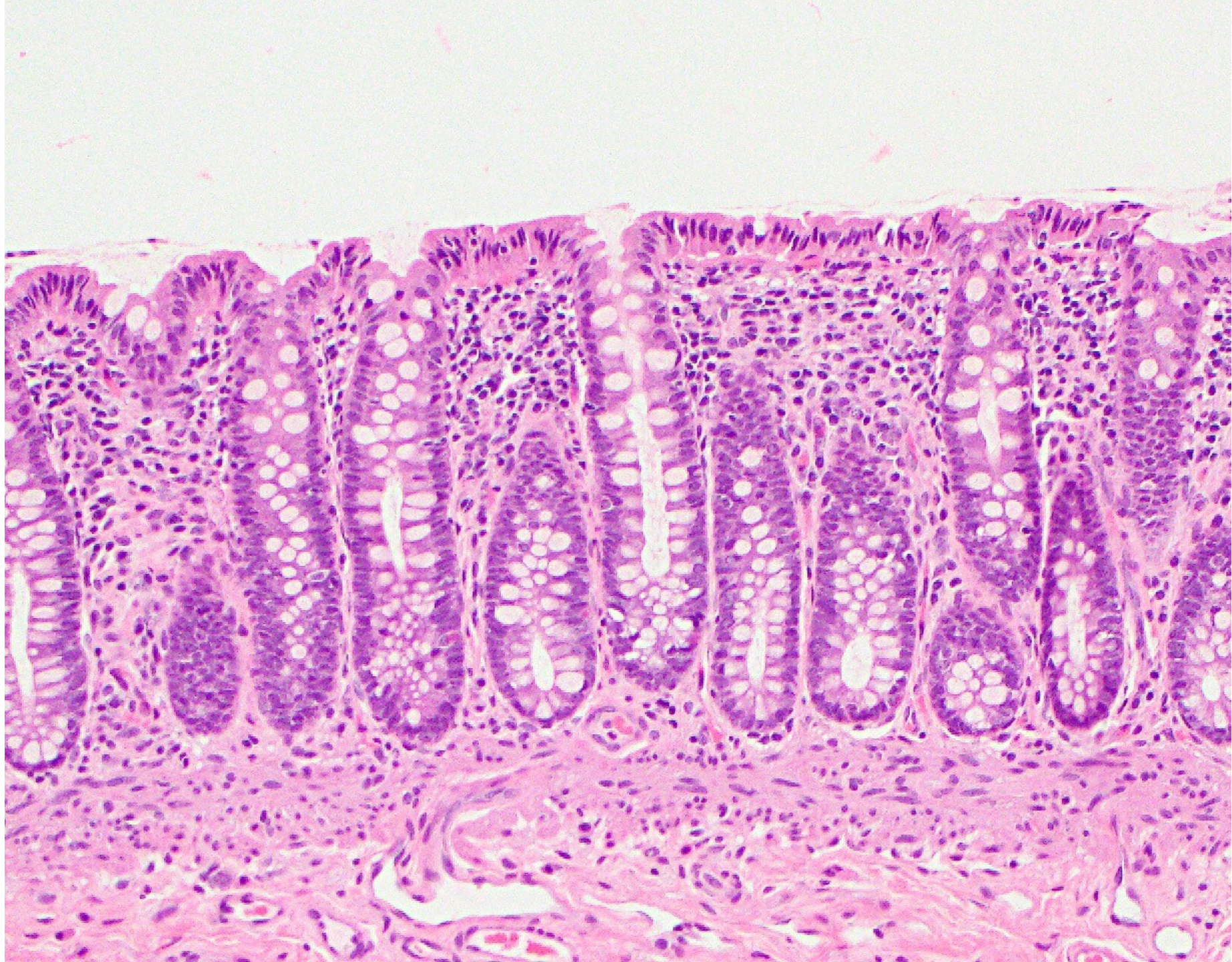
Necrotizing enterocolitis (NEC)

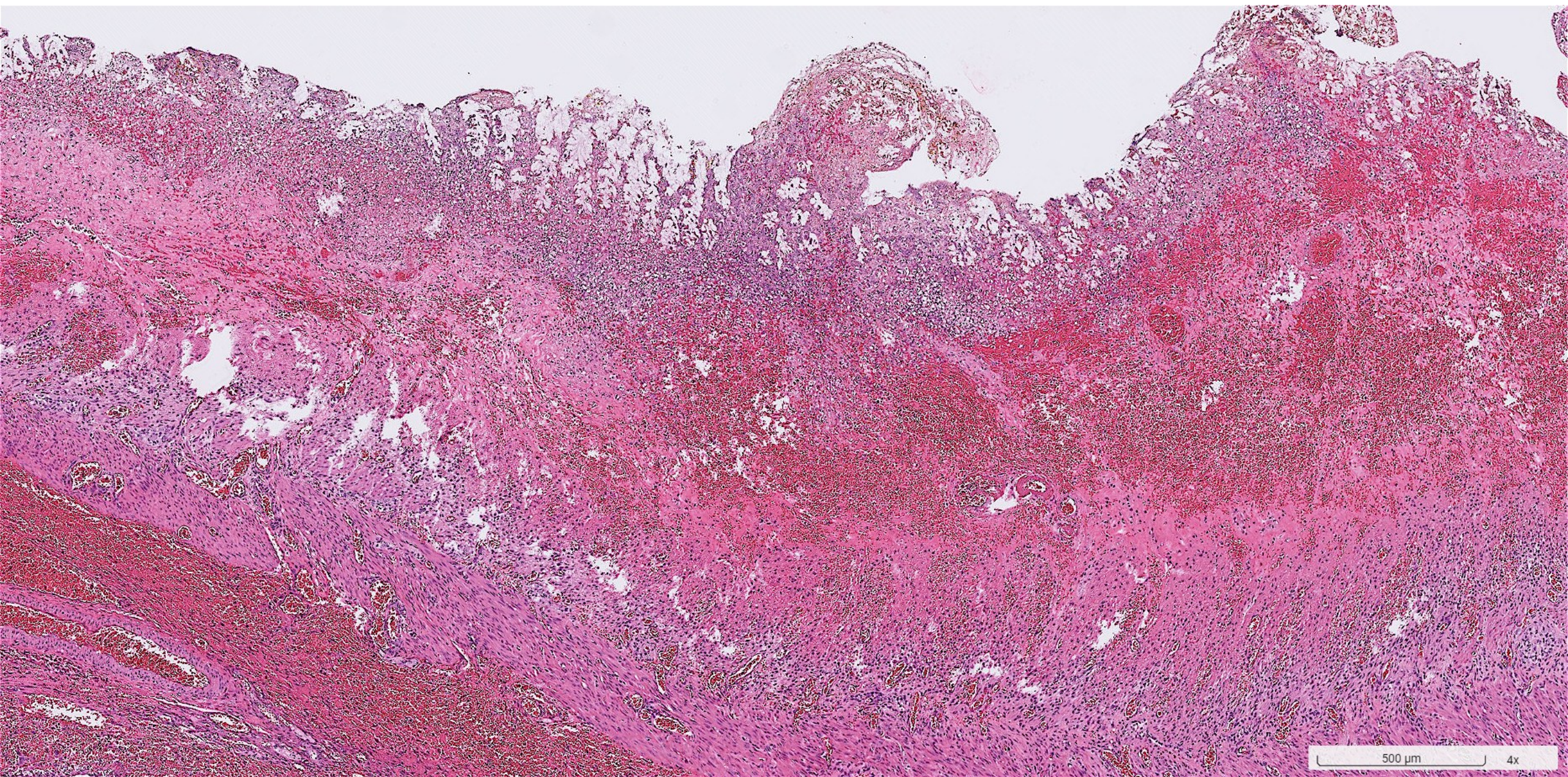
- Idiopathic enterocolitis: Coagulative/hemorrhagic necrosis and inflammation of small bowel and proximal colon
- Most often in premature infants; very high mortality
- Etiology/Pathogenesis:
 - Premature birth: Intestinal immaturity
 - Microbial dysbiosis (abnormal intestinal microbiota)
 - Enteral feeding: Breast milk and probiotics prevent
 - Intestinal mucosal ischemia (ischemic enterocolitis)

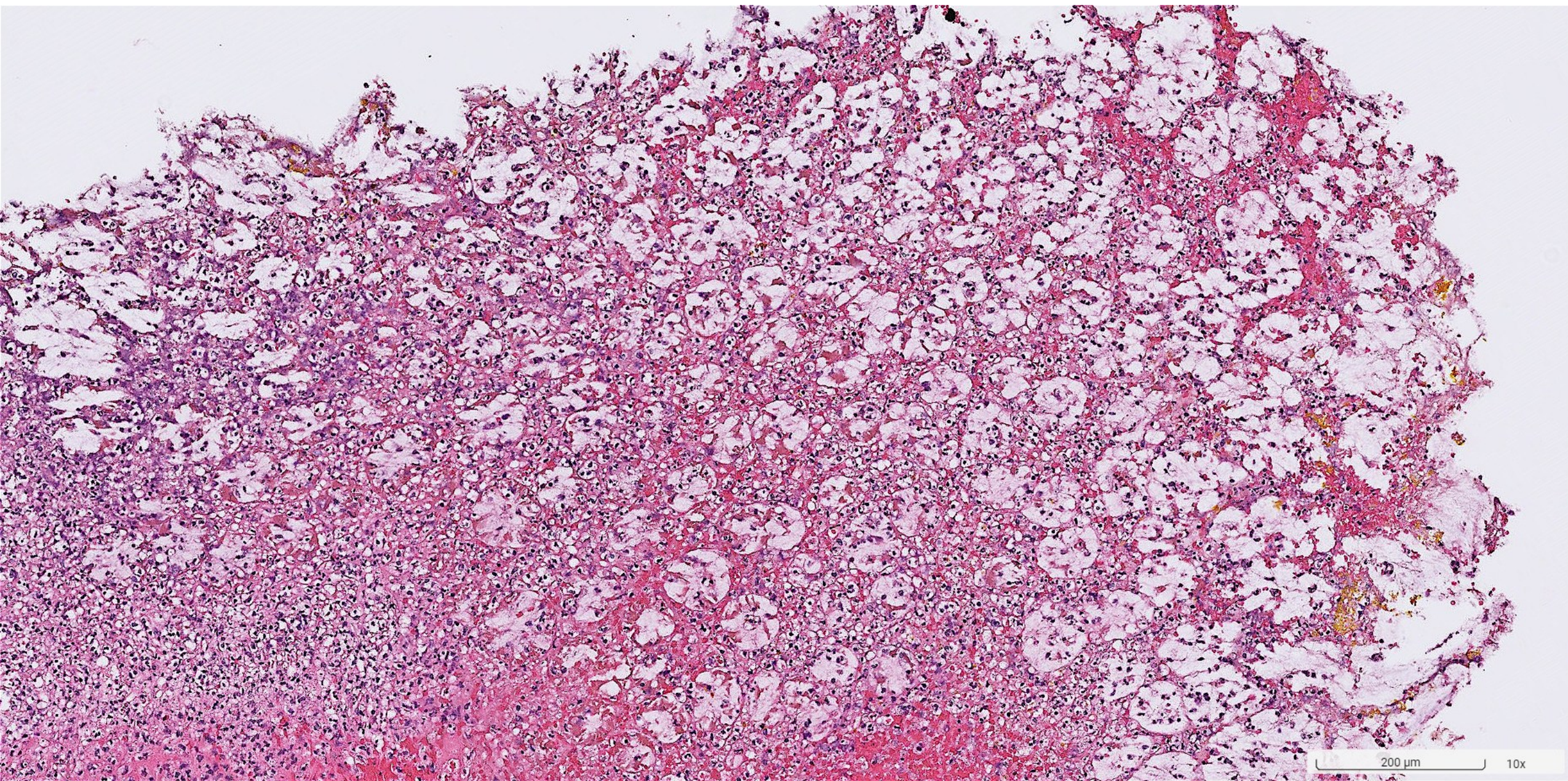
Colon (low power):

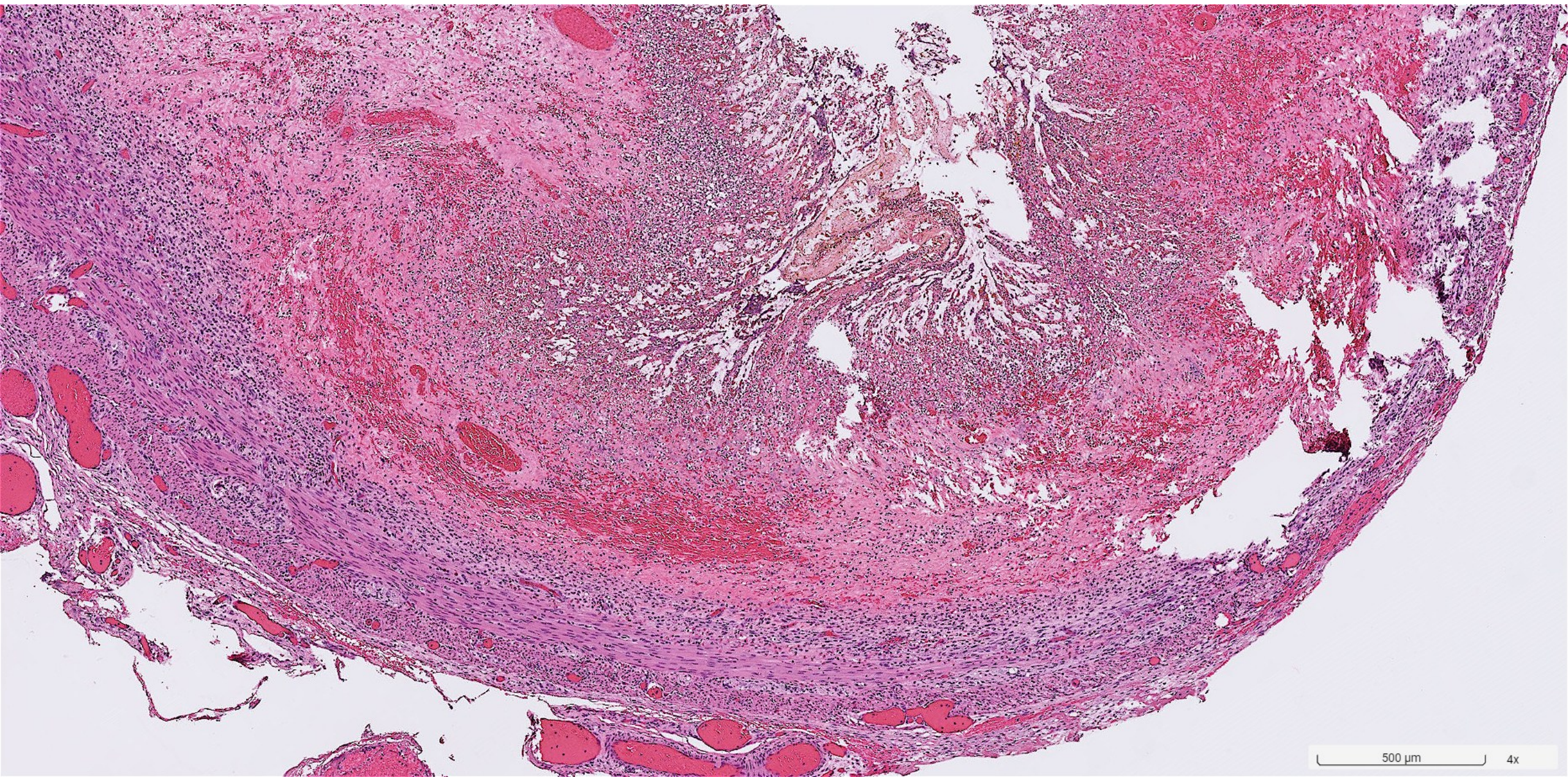
A section of the colon. The mucosa contains crypts but no villi.





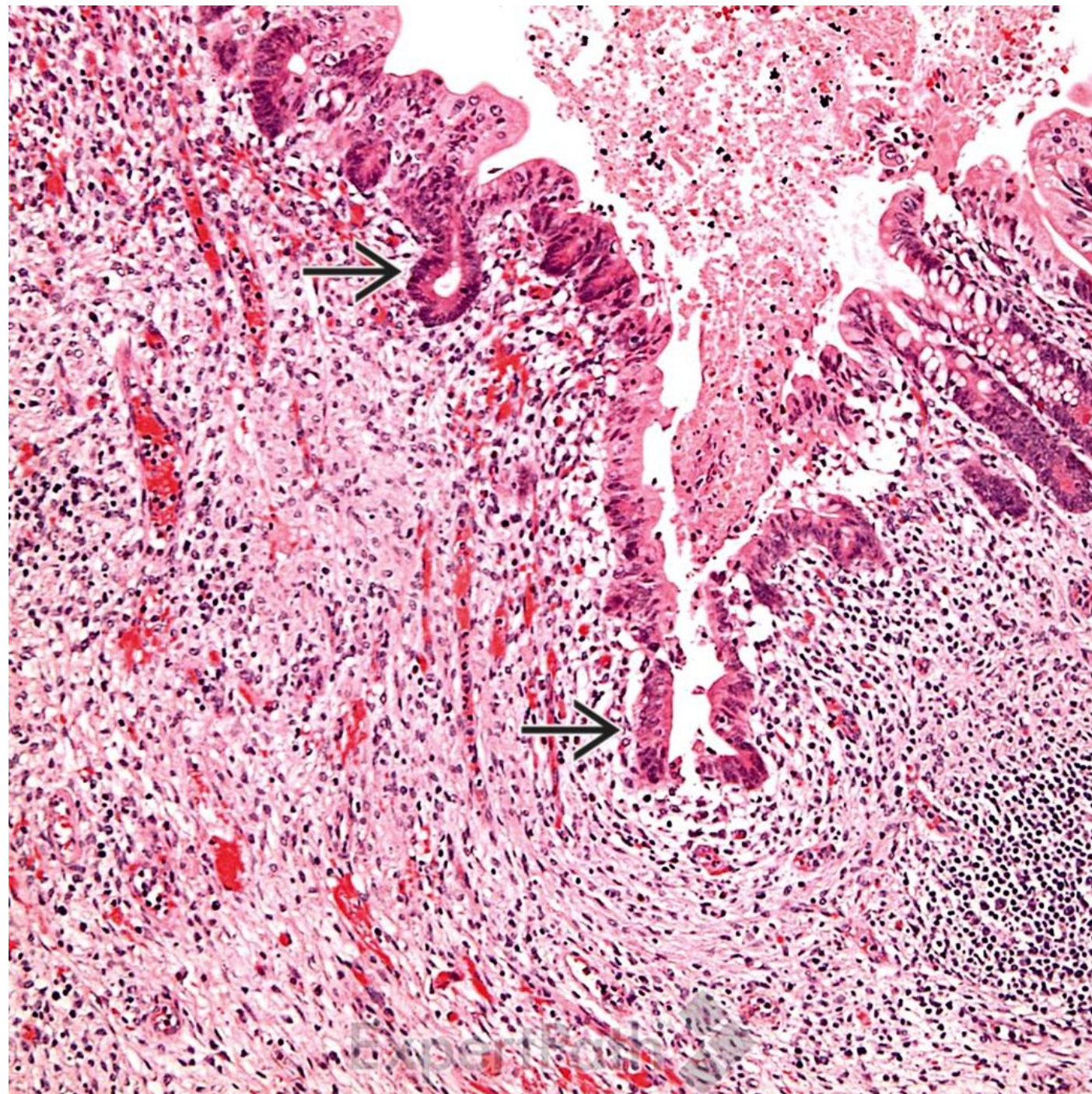






500 μ m 4x

器官 (Organ)	Intestine/colon
診斷 (Diagnosis)	Necrotizing enterocolitis
描述 (Description)	Coagulative necrosis with hemorrhage – mucosal and transmural (impending perforation) - Chronic changes: strictures, crypt distortion, reactive epithelial cells (本片無)



Take Home Message

- Cell injury → Necrosis
- **Coagulative necrosis**
 - Eosinophilic dead cells having **preserved cellular outlines**, without nuclei → “Ghost cells”
 - Caused by ischemia
- Review histology → Recognize **normal vs abnormal**